

Technology Stack

Date	26 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js 18+, Vite, Lucide React Icons, Recharts	Modern SPA framework with fast hot-reload (Vite); Recharts for financial data visualization; Lucide for consistent iconography

2	Backend / Server-Side	Python 3.10+, FastAPI	High-performance async web framework with automatic OpenAPI docs; native Pydantic validation for financial data schemas
3	Database / Data Storage	SQLite (via SQLAlchemy ORM)	Lightweight relational database ideal for local deployment; SQLAlchemy provides clean ORM patterns for complex financial relationships
4	AI / Machine Learning	Google Gemini API (gemini-1.5-flash)	State-of-the-art generative AI for producing personalized, context-aware negotiation letters and strategies

5	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Development Server (Uvicorn + Vite Dev Server)	Cost-effective for demonstration; Uvicorn provides ASGI server for FastAPI; easily deployable on any machine
6	Version Control & CI/CD	Git, GitHub	Seamless team collaboration, version tracking, and code review; free hosting for student projects